

Python Type Analyzer

RQ1
(Analysis Approach)

Static (w/o runtime informations)

Dynamic (w/ runtime informations)

Hybrid(Static + Dynamic)

RQ2
(Analysis Features)

Structural Typing

Flow-sensitivity

Dynamic Features

Compound DataTypes

RQ3
(Existing tools)

Static Analyzer

Mypy, pyright ...

Dynamic Checking

Typeguard, MonkeyType ...

Hybrid Analyzer

HiTyper, CrossHair ...

LLM-based Inference

TypeWriter, Type4Py ...